Algebraic Geometry

Ch.3: Coordinate Rings (week 4)

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Outline

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The affine dictionary we want

$$X \longmapsto K[X] \longmapsto \mathrm{Spec}(K[X]).$$

- Start with an affine algebraic set $X \subseteq \mathbb{A}^n_K$.
- Pass to its intrinsic algebra of polynomial functions $K[X]$.
- Recover classical points via maximal ideals.
- Enlarge to all prime ideals to capture irreducible closed subsets.
- Use localization to describe functions on basic opens $D(f)$.

$$\mathrm{AffAlgSet}_K^{\mathrm{op}} \simeq \mathrm{Alg}_{K,\mathrm{fg},\mathrm{red}}, \qquad \mathrm{AffSch}^{\mathrm{op}} \simeq \mathrm{CRing}.$$

Three affine levels

- **Classical reducible geometry:**

$$\mathrm{AffAlgSet}_K \leftrightarrow \mathrm{Alg}_{K,\mathrm{fg},\mathrm{red}}.$$

- **Classical irreducible geometry:**

$$\mathrm{AffVar}_K \leftrightarrow \mathrm{Alg}_{K,\mathrm{fg},\mathrm{dom}}.$$

- **Affine schemes over arbitrary rings:**

$$\mathrm{AffSch}^{\mathrm{op}} \simeq \mathrm{CRing}.$$

- Only the first two need K algebraically closed; the third is the structural framework behind both.

Why functions come first

- An embedding $X \hookrightarrow \mathbb{A}^n_K$ is not canonical.
- A generating set of equations for X is not canonical either.
- The ring of polynomial functions on X is intrinsic.
- Coordinate rings package geometry as algebra: addition, multiplication, and algebraic relations among functions.

Polynomial algebras are free coordinate algebras

Proposition (Free commutative K-algebra on n generators)

For every commutative K-algebra A and every $(a_1, \ldots, a_n) \in A^n$, there is a unique K-algebra map

$$\mathrm{ev}_{(a_1,\ldots,a_n)} : K[x_1, \ldots, x_n] \longrightarrow A$$

with $x_i \mapsto a_i$.

$$\boxed{\mathrm{Hom}_{K\text{-alg}}(K[x_1, \ldots, x_n], A) \cong A^n.}$$

$$\begin{tikzcd}
K \arrow[r, "\iota"] \arrow[dr, "\eta_A"'] & K[x_1, \ldots, x_n] \arrow[d, dashed, "\exists!\, \mathrm{ev}_{(a_1,\ldots,a_n)}"] \\
& A
\end{tikzcd}$$

Polynomial functions and the coordinate ring

Definition (Polynomial function)

A polynomial function on $X \subseteq \mathbb{A}^n_K$ is a function $X(K) \to K$ obtained by restricting some polynomial in $K[x_1, \ldots, x_n]$.

Definition (Coordinate ring)

$$K[X] := \{f|_X : f \in K[x_1, \ldots, x_n]\},$$

with pointwise ring operations.

- Different ambient polynomials may define the same function on X.
- The ideal of relations is
$$I(X) = \{f \in K[x_1, \ldots, x_n] : f|_X = 0\}.$$

Quotient rings impose relations universally

Proposition (Universal property of the quotient)

Let $q : R \to R/I$ be the quotient map. For every ring map $\varphi : R \to A$ with $I \subseteq \ker \varphi$, there is a unique $\overline{\varphi} : R/I \to A$ with $\varphi = \overline{\varphi} \circ q$.

$$\begin{tikzcd} R \arrow[r, "q"] \arrow[dr, "\varphi"'] & R/I \arrow[d, dashed, "\exists!\,\overline{\varphi}"] \\ & A \end{tikzcd}$$

- R/I is initial among R-algebras on which every element of I acts by zero.
- Equivalently: every ring map annihilating I factors uniquely through R/I.

Coordinate rings as universal quotients

Corollary (Coordinate ring as universal quotient)

For $X \subseteq \mathbb{A}^n_K$,

$$\boxed{K[X] \cong K[x_1, \ldots, x_n]/I(X).}$$

$$\begin{tikzcd}
K[x_1, \ldots, x_n] \arrow[r, "\rho"] \arrow[d, "q_X"'] & K[X] \\
K[x_1, \ldots, x_n]/I(X) \arrow[ur, "\sim"'] &
\end{tikzcd}$$

- Quotienting by $I(X)$ identifies two polynomials precisely when they induce the same function on X.
- Therefore $K[X]$ is finitely generated and reduced in the classical setting.

Running example: the parabola

Let

$$X = V(y - x^2) \subseteq \mathbb{A}^2_K.$$

Then

$$K[X] \cong K[x, y]/(y - x^2) \cong K[x].$$

- Inside the quotient, y becomes x^2.
- So the parabola has the same coordinate ring as the affine line.
- Geometrically: the chosen equations may differ, but the coordinate ring captures the intrinsic affine geometry.

$$\boxed{K[V(y - x^2)] \cong K[x].}$$

Classical points as evaluation maps

For each $x \in X(K)$, evaluation gives a K-algebra map

$$\mathrm{ev}_x : K[X] \longrightarrow K, \qquad f \longmapsto f(x).$$

Its kernel

$$\mathfrak{m}_x := \ker(\mathrm{ev}_x)$$

is a maximal ideal because constants lie in $K[X]$, so the map is surjective and

$$K[X]/\mathfrak{m}_x \cong K.$$

$$\begin{tikzcd}
K[X] \arrow[r, "\mathrm{ev}_x"] \arrow[d, "q_x"'] & K \\
K[X]/\mathfrak{m}_x \arrow[ur, "\sim"'] &
\end{tikzcd}$$

Nullstellensatz gives the classical dictionary

Theorem (Hilbert's Nullstellensatz)

If K is algebraically closed, then:

1. $I(V(I)) = \mathrm{rad}(I)$ *for every ideal* $I \subseteq K[x_1, \ldots, x_n]$;
2. *maximal ideals of* $K[x_1, \ldots, x_n]$ *are exactly*

$$(x_1 - a_1, \ldots, x_n - a_n), \qquad (a_1, \ldots, a_n) \in K^n.$$

Hence

$$\boxed{X(K) \cong \mathrm{Hom}_{K\text{-alg}}(K[X], K) \cong \mathrm{MaxSpec}(K[X]).}$$

Example: the affine line inside its spectrum

Take $R = K[x]$.

- Closed points are the maximal ideals $(x - a)$ for $a \in K$.
- Thus
$$\mathrm{MaxSpec}(K[x]) \cong \mathbb{A}^1_K(K).$$
- But $\mathrm{Spec}(K[x])$ has one more prime ideal: (0).
- Its closure is the whole space:
$$\overline{\{(0)\}} = \mathrm{Spec}(K[x]).$$

This is our first generic point.

Why maximal ideals are not enough

- $\mathrm{MaxSpec}(R)$ recovers classical closed points.
- But irreducible closed subsets usually do not have points in MaxSpec representing them.
- Passing to

$$\mathrm{Spec}(R) := \{\mathfrak{p} \subseteq R : \mathfrak{p} \text{ prime}\}$$

adds the generic points needed to encode irreducible geometry.

$$\boxed{X(K) \;\cong\; \mathrm{MaxSpec}(K[X]) \;\subseteq\; \mathrm{Spec}(K[X]).}$$

Zariski topology on the spectrum

For an ideal $I \subseteq R$, define

$$V(I) := \{\mathfrak{p} \in \operatorname{Spec}(R) : I \subseteq \mathfrak{p}\}.$$

For $f \in R$, define the standard open

$$D(f) := \operatorname{Spec}(R) \setminus V((f)) = \{\mathfrak{p} : f \notin \mathfrak{p}\}.$$

- $D(1) = \operatorname{Spec}(R), \quad D(0) = \varnothing.$
- $D(fg) = D(f) \cap D(g)$, since $fg \notin \mathfrak{p} \iff f \notin \mathfrak{p}$ and $g \notin \mathfrak{p}$ *(primality)*.

So the sets $D(f)$ form a basis for the Zariski topology.

Prime ideals correspond to irreducible closed sets

For $\mathfrak{p} \in \operatorname{Spec}(R)$,

$$\boxed{\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).}$$

Therefore:

- $\mathfrak{p}$ is closed iff $\mathfrak{p}$ is maximal;
- prime ideals are in bijection with irreducible closed subsets;
- every irreducible closed subset has a unique generic point.

$$\boxed{\{\text{prime ideals of } R\} \longleftrightarrow \{\text{irreducible closed subsets of } \operatorname{Spec}(R)\}.}$$

Example: generic point of the parabola

In $R = K[x, y]$, let

$$\mathfrak{p} = (y - x^2).$$

Since

$$R/\mathfrak{p} \cong K[x]$$

is a domain, $\mathfrak{p}$ is prime. Hence

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

- $\mathfrak{p}$ is the generic point of the parabola.
- Every closed point $(x - a, y - a^2)$ is a specialization:

$$\mathfrak{p} \subseteq (x - a,\ y - a^2).$$

Affine plane: three levels of points

In $\mathrm{Spec}(K[x,y])$:

- **Closed points:** maximal ideals $(x-a, y-b)$, i.e. classical points $(a,b) \in K^2$.
- **Curve generic points:** e.g. $(y-3x^2)$ is the generic point of $V(y-3x^2)$, with

$$\overline{\{(y-3x^2)\}} = V(y-3x^2).$$

- **Ambient generic point:** (0) with

$$\overline{\{(0)\}} = \mathrm{Spec}(K[x,y]).$$

$$\boxed{\text{classical points} \leftrightarrow \text{maximal ideals} \subsetneq \text{all prime ideals.}}$$

Localization is universal inversion

Let $S \subseteq R$ be multiplicative. The localization map

$$i_S : R \longrightarrow S^{-1}R$$

is characterized by:

- every $s \in S$ becomes invertible in $S^{-1}R$;
- for any ring map $\varphi : R \to A$ with $\varphi(S) \subseteq A^\times$, there is a unique factorization:

$$\begin{tikzcd} R \arrow[r, "i_f"] \arrow[dr, "\varphi"'] & R_f \arrow[d, dashed, "\exists!\ \widetilde{\varphi}"] \\ & A \end{tikzcd}$$

In particular, R_f is the universal R-algebra in which f becomes invertible.

Localization matches basic opens

For $f \in R$, the localization map $i_f : R \to R_f$ induces a homeomorphism

$$\mathrm{Spec}(R_f) \xrightarrow{\sim} D(f), \qquad \mathfrak{q} \longmapsto i_f^{-1}(\mathfrak{q}).$$

- Prime ideals of R_f correspond to prime ideals of R disjoint from $\{1, f, f^2, \dots\}$.
- Equivalently, they correspond exactly to primes not containing f.

$$\boxed{\mathrm{Spec}(R_f) \cong D(f).}$$

The structure sheaf on an affine spectrum

For $X = \mathrm{Spec}(R)$, define the structure sheaf on basic opens by

$$\mathcal{O}_X(D(f)) := R_f.$$

So regular functions on $D(f)$ are exactly fractions

$$\frac{a}{f^n}, \qquad a \in R,\ n \geq 0.$$

Key slogan

Localization is the algebra behind restricting functions to the open set where f does not vanish.

Example: the punctured affine line

Let $R = K[x]$ and $X = \mathrm{Spec}(K[x])$.

$$D(x) = \{\mathfrak{p} \in \mathrm{Spec}(K[x]) : x \notin \mathfrak{p}\}.$$

Then

$$\Gamma(D(x), \mathcal{O}_X) = K[x]_x = K[x, x^{-1}].$$

- x^{-1} is regular on $D(x)$;
- but x^{-1} is not global on all of X;
- indeed $\Gamma(X, \mathcal{O}_X) = K[x]$.

The classical affine dictionary

Theorem (Classical affine dictionary)

Over algebraically closed K, there is an anti-equivalence of categories

$$\boxed{\mathrm{AffAlgSet}_K^{\mathrm{op}} \simeq \mathrm{Alg}_{K,\mathrm{fg},\mathrm{red}}.}$$

Restricting to irreducible objects refines to

$$\mathrm{AffVar}_K^{\mathrm{op}} \simeq \mathrm{Alg}_{K,\mathrm{fg},\mathrm{dom}}.$$

- The geometric direction sends $X \mapsto K[X]$.
- The algebraic direction sends $A \mapsto \mathrm{MaxSpec}(A)$.
- Arrow reversal (anti-equivalence) comes from pullback of functions.

Example: recovering equations from a ring

Let

$$A := K[u^2, uv, v^2] \subseteq K[u, v].$$

Define a surjective map

$$\psi : K[x, y, z] \twoheadrightarrow A, \qquad x \mapsto u^2,\ y \mapsto uv,\ z \mapsto v^2.$$

- The relation $xz - y^2 = 0$ holds in A, and in fact

$$\ker \psi = (xz - y^2).$$

- So $A \cong K[x, y, z]/(xz - y^2)$, and the associated affine algebraic set is

$$V(xz - y^2) \subseteq \mathbb{A}^3_K$$

(a quadratic cone).

Morphisms reverse arrows on functions

A morphism $f : X \to Y$ induces pullback of functions

$$f^* : K[Y] \longrightarrow K[X], \qquad g \longmapsto g \circ f.$$

$$\begin{tikzcd}
K[Y] \arrow[r, "f^*"] \arrow[d, "\mathrm{ev}_{f(x)}"'] & K[X] \arrow[d, "\mathrm{ev}_x"] \\
K \arrow[r, equal] & K
\end{tikzcd}$$

- The square commutes: evaluating $f^*(g)$ at x equals evaluating g at $f(x)$.
- This arrow reversal is why the coordinate-ring construction is contravariant.

Ring maps induce maps of spectra

Proposition

A ring homomorphism $\varphi : R \to S$ induces a continuous map

$$\mathrm{Spec}(S) \longrightarrow \mathrm{Spec}(R), \qquad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q}).$$

- Preimage of a prime ideal under a ring map is always prime.
- Continuity: φ^{-1} sends $V(I) \subseteq \mathrm{Spec}(R)$ to $V(I \cdot S) \subseteq \mathrm{Spec}(S)$.
- This is the geometric shadow of the algebraic map φ.

Universal property of Spec

Theorem (Universal property of affine target)

For every locally ringed space X and commutative ring R, there is a natural bijection

$$\boxed{\mathrm{Hom}_{\mathrm{LRS}}(X, \mathrm{Spec}(R)) \cong \mathrm{Hom}_{\mathrm{CRing}}(R, \Gamma(X, \mathcal{O}_X)).}$$

$$\begin{tikzcd}
R \arrow[r, "\alpha"] \arrow[d, "\cong"'] & \Gamma(X, \mathcal{O}_X) \\
\Gamma(\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)}) \arrow[ur, "\Gamma(u_\alpha)"'] &
\end{tikzcd}$$

- A morphism into an affine scheme is determined by a ring map on global sections.
- $\mathrm{Spec}(R)$ *represents* the functor $X \mapsto \mathrm{Hom}_{\mathrm{CRing}}(R, \Gamma(X, \mathcal{O}_X))$.

Affine Hom-bijections

Theorem (Classical affine Hom-bijection)

For affine algebraic sets X, Y over algebraically closed K,

$$\boxed{\mathrm{Hom}_{\mathrm{AffAlgSet}_K}(X, Y) \cong \mathrm{Hom}_{K\text{-}alg}(K[Y], K[X]).}$$

Theorem (Affine schemes Hom-bijection)

For commutative rings R, S,

$$\boxed{\mathrm{Hom}_{\mathrm{AffSch}}(\mathrm{Spec}(S), \mathrm{Spec}(R)) \cong \mathrm{Hom}_{\mathrm{CRing}}(R, S).}$$

- The scheme version uses $\Gamma(\mathrm{Spec}(S), \mathcal{O}_{\mathrm{Spec}(S)}) \cong S$.
- Geometry and algebra determine each other; the ring map always points in the opposite direction.

Products and fibre products

Proposition

Affine schemes admit finite limits. In particular,

$$\boxed{\mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(B) \cong \mathrm{Spec}(A \otimes_R B).}$$

$$\begin{tikzcd}
\mathrm{Spec}(A \otimes_R B) \arrow[r, "\mathrm{pr}_B"] \arrow[d, "\mathrm{pr}_A"'] & \mathrm{Spec}(B) \arrow[d] \\
\mathrm{Spec}(A) \arrow[r] & \mathrm{Spec}(R)
\end{tikzcd}$$

- Tensor products encode affine base change on spectra.
- **Functor of points:** for $\mathrm{Spec}(R)$,

$$h_{\mathrm{Spec}(R)}(A) := \mathrm{Hom}_{\mathrm{AffSch}}(\mathrm{Spec}(A), \mathrm{Spec}(R)) \cong \mathrm{Hom}_{\mathrm{CRing}}(R, A).$$

- The ring R is recovered from its functor of points.

The big picture

$$X \longmapsto K[X] \longmapsto \mathrm{Spec}(K[X]).$$

1. **Coordinate rings** encode polynomial functions intrinsically.
2. **Maximal ideals** recover classical points via evaluation.
3. **Prime ideals** add generic points for irreducible closed subsets.
4. **Localization** produces functions on $D(f)$; $\mathrm{Spec}(R_f) \cong D(f)$.
5. **Contravariance** reverses arrows: morphisms of spaces $\longleftrightarrow$ ring maps in the opposite direction.

The organizing principle is **functorial**:

$$\boxed{\mathrm{AffAlgSet}_K^{\mathrm{op}} \simeq \mathrm{Alg}_{K,\mathrm{fg},\mathrm{red}}, \qquad \mathrm{AffSch}^{\mathrm{op}} \simeq \mathrm{CRing}.}$$